Kairos: from observed history to executable progress

A single ready order, a separate capacity transaction, and layout-aware state continuity

Causal state snapshot

Arrived prompt | Completed history | Actual KV layout | Ready / running / tool

1 Information

Text + history → Encoder

Current output
P(O | history)

Conditional service
P(R+ | O, history)

Infer once per arrived call.
Condition on observed output survival.

2 Program scheduling

Geometric J = exp(E[log(t / g)])

Ready order: **A** **B** **C** **D**

Increasing J; cold and returning calls together

Independent capacity transaction

Native-frontier pages + execution slots
Earlier claim + eligible optional victims

4 Restore & execute

GPU pages

CPU chunks

Use the joint recoverable prefix.
Copy missing pages; recompute gaps.

Native allocator + vLLM / LMCache

• Program & tool

LLM → tool wait → next arrived call

3 KV continuity

Keep benefit: actual recovery saved

Return probability × benefit − pressure × hold time

Feedback contains completed observations only. Solid: decision / execution flow. Dashed: completed-state feedback.